

- 8 (a) Explain what is meant by the term *biodiversity*.

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- (b) Simpson's Index of Diversity (D) is a method of assessing biodiversity.

The formula for Simpson's Index of Diversity is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

n = number of individuals of each species present in the sample

N = total number of all individuals of all species

- (i) In an investigation of biodiversity in a pond, samples of pond animals were removed using sampling nets. The species of each animal was identified and the number of individuals of each species was recorded.

Table 8.1 shows the results of the investigation.

Calculate Simpson's Index of Diversity by completing Table 8.1 in the spaces provided. Record your values to three decimal places.

Write the value for Simpson's Index of Diversity on the dotted line. Record your value to three decimal places.

Table 8.1

species	number	$\frac{n}{N}$	$\left(\frac{n}{N} \right)^2$
<i>Rana temporaria</i>	10	0.042	0.002
<i>Leucorrhinia dubia</i>	35	0.148	0.022
<i>Hydrometra stagnorum</i>	50	0.212	0.045
<i>Lymnaea stagnalis</i>	44	0.186	0.035
<i>Gammarus pulex</i>	97		
total	236		

Simpson's Index of Diversity (D) =

[3]

- (ii) Explain what this value for Simpson's Index of Diversity shows about the diversity of the pond.

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[Total: 8]

Section B

Answer **one** question.